

1 (a) Define *work done*.

.....
 [1]

(b) A trolley of mass 400 g is moving at a constant speed of 2.5 m s^{-1} to the right as shown in Fig. 1.1.

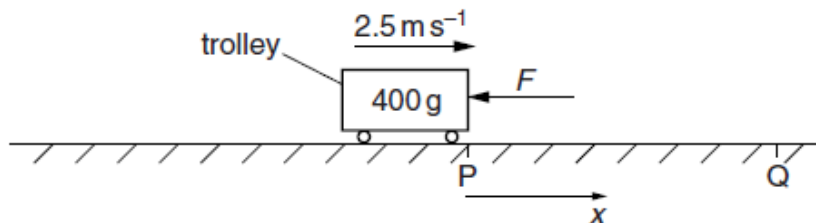


Fig. 1.1

A variable force F acts to the left on the trolley as it moves between points P and Q. The variation of F with displacement x from P is shown in Fig. 1.2.

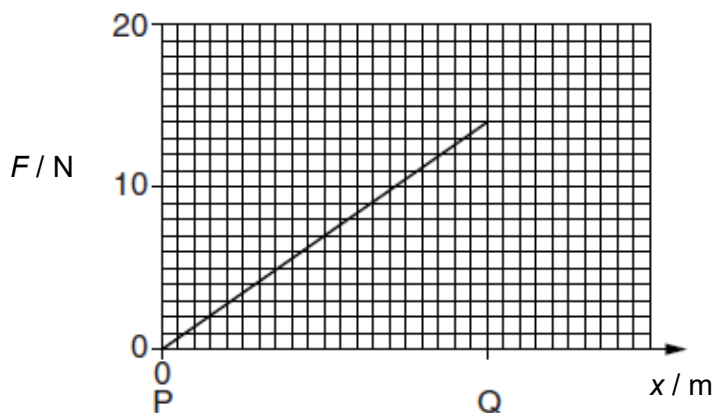


Fig. 1.2

The trolley comes to rest at point Q.

(i) Calculate the distance PQ.

distance = m [2]

- (ii) On Fig. 1.3, sketch, with appropriate values, the variation with x of work done on trolley by F .

work done by F / J



Fig. 1.3

[3]

- (iii) In order to maintain a constant speed of 2.5 m s^{-1} , an electric motor attached to the trolley is switched on.

On Fig. 1.4, sketch the variation with x of the power supplied by motor while the trolley moves from point P to Q. No numerical value is required.

power / W



Fig. 1.4

[1]